ROB 510 Midterm Sheet

MLS

conventions, $\xi = (v, \omega)$

Rotations: SO(3)

Columns: $R_{ab} = [x_{ab}\ y_{ab}\ z_{ab}]$ = B's axes in A coords. $q_a = R_{ab} q_b$. $R_{ac} = R_{ab} R_{bc}$, $R_{ba} = R_{ab}^T$. Translate words to subscripts and chain inner letters.

$SO(3) = \{RR^T = I,\ \det R = +1\}$ (det $+1$ excludes reflections). RBT = preserves length + cross product (hence inner product).

Groups: closure, identity, inverse, associativity. Abelian: commutes. SO(2) abelian; SO($n \geq 3$), SE(3) not. so(3): abelian group under *addition* (vector space), not under multiplication.

Hat: $\hat{a} = \begin{bmatrix} 0 & -a_3 & a_2 \\ a_3 & 0 & -a_1 \\ -a_2 & a_1 & 0 \end{bmatrix}$, $\hat{a} b = a \times b$,

$\hat{a}^T = -\hat{a}$. $\widehat{Rw} = R\hat{w}R^T$. $\hat{w}w = 0 \Rightarrow \det \hat{w} = 0$ **always**. Eigs of $\hat{\omega}$ ($\|\omega\|$=1): $0, \pm i$; eigvec of 0 is ω.

$\hat{a}^2 = aa^T - \|a\|^2 I$, $\hat{a}^3 = -\|a\|^2 \hat{a}$. Unit ω: $\hat{\omega}^2 = \omega\omega^T - I$, $\hat{\omega}^3 = -\hat{\omega}$.

Cross: $x \times y = z$, $y \times z = x$, $z \times x = y$; reverse order gives $-$.

$a \times b = (a_2 b_3 - a_3 b_2,\ a_3 b_1 - a_1 b_3,\ a_1 b_2 - a_2 b_1)$.

Elementary: $R_x(\phi) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & c & -s \\ 0 & s & c \end{bmatrix}$

$R_y = \begin{bmatrix} c & 0 & s \\ 0 & 1 & 0 \\ -s & 0 & c \end{bmatrix}$ $R_z = \begin{bmatrix} c & -s & 0 \\ s & c & 0 \\ 0 & 0 & 1 \end{bmatrix}$

Rodrigues (**needs $\|\omega\| = 1$**):

$e^{\hat{\omega}\theta} = I + \hat{\omega}\sin\theta + \hat{\omega}^2(1 - \cos\theta)$. Proof: odd powers $(-1)^{n-1}\theta^{2n-1}\hat{\omega}$, even $(-1)^{n+1}\theta^{2n}\hat{\omega}^2$, so the series give sin and $1 - \cos$.

Non-unit ω: use $\hat{\omega}\theta = (\widehat{\omega/\|\omega\|})(\|\omega\|\theta)$.

Coordinate axis: $e^{\hat{z}\theta} = R_z(\theta)$.

exp∈SO(3) (P2.4):

$(e^{\hat{\omega}\theta})^{-1} = e^{-\hat{\omega}\theta} = (e^{\hat{\omega}\theta})^T$; det=+1 by continuity from I.

Log / surjective (P2.5, Euler Thm 2.6):

$\theta = \cos^{-1}\frac{\mathrm{tr}R - 1}{2}$;

$\omega = \frac{1}{2\sin\theta}(r_{32} - r_{23},\ r_{13} - r_{31},\ r_{21} - r_{12})$.

- $\theta = 0$ ($R = I$): **singular**, ω arbitrary (infinitely many exp coords).
- $\theta = \pi$: not singular. $R = 2\omega\omega^T - I$, so $\omega_i = \pm\sqrt{(r_{ii} + 1)/2}$, relative signs from $r_{ij} = 2\omega_i\omega_j$.
- Same R: $(\omega, \theta) \sim (-\omega, -\theta) \sim (-\omega, 2\pi - \theta) \sim (\omega, \theta + 2\pi k)$.

exp surjective, NOT injective.

Eigs of R: $1, e^{\pm i\theta}$ (same eigvecs as $\hat{\omega}$); eigvec for 1 = axis. Orthogonal $A \in SO(3) \iff \prod \lambda = +1$.

SO(2) $\cong S^1$: $e^{\hat{\omega}\theta} = R(\omega\theta)$; exp surjective, not injective; $R\hat{\omega}R^T = \hat{\omega}$ in 2D only.

Euler angles (local, 3 params)

Intrinsic (new axes) = post-multiply; extrinsic (fixed axes) = pre-multiply (reverse order).

ZYZ $R_z(\alpha)R_y(\beta)R_z(\gamma) =$

$$\begin{bmatrix} c_\alpha c_\beta c_\gamma - s_\alpha s_\gamma & -c_\alpha c_\beta s_\gamma - s_\alpha c_\gamma & c_\alpha s_\beta \\ s_\alpha c_\beta c_\gamma + c_\alpha s_\gamma & -s_\alpha c_\beta s_\gamma + c_\alpha c_\gamma & s_\alpha s_\beta \\ -s_\beta c_\gamma & s_\beta s_\gamma & c_\beta \end{bmatrix}$$

$\beta = \mathrm{atan2}(\sqrt{r_{31}^2 + r_{32}^2}, r_{33})$,

$\alpha = \mathrm{atan2}(r_{23}, r_{13})$, $\gamma = \mathrm{atan2}(r_{32}, -r_{31})$ (divide by s_β).

Singular at $\beta = 0, \pi$: $R = I \Rightarrow (\alpha, 0, -\alpha)$, infinitely many.

ZYX $R_z(\phi)R_y(\theta)R_x(\psi)$: col 1 $= (c_\phi c_\theta, s_\phi c_\theta, -s_\theta)$, row 3 $= (-s_\theta, c_\theta s_\psi, c_\theta c_\psi)$.

$\theta = \mathrm{atan2}(-r_{31}, \sqrt{r_{11}^2 + r_{21}^2})$,

$\phi = \mathrm{atan2}(r_{21}, r_{11})$, $\psi = \mathrm{atan2}(r_{32}, r_{33})$.

Singular at $\theta = \pm\pi/2$.

No 3-param rep of SO(3) is singularity-free.

Euler→R is **not injective**. Quaternions (4) are global.

SE(3), homogeneous coords

$g = \left[\begin{smallmatrix} R & p \\ 0 & 1 \end{smallmatrix}\right]$, $\bar{q} = (q, 1)$ point, $\bar{v} = (v, 0)$ vector (translation ignored: $\bar{g}\bar{v} = (Rv, 0)$).

$g^{-1} = \left[\begin{smallmatrix} R^T & -R^T p \\ 0 & 1 \end{smallmatrix}\right]$,

$g_{ac} = g_{ab} g_{bc} = \left[\begin{smallmatrix} R_{ab}R_{bc} & R_{ab}p_{bc} + p_{ab} \\ 0 & 1 \end{smallmatrix}\right]$. $\det g = 1$ (even pure translation).

pt−pt = vec; pt+vec = pt; pt+pt meaningless (bottom 2).

Twists, exp map on SE(3)

$\hat{\xi} = \left[\begin{smallmatrix} \hat{\omega} & v \\ 0 & 0 \end{smallmatrix}\right]$ (**bottom row all 0**),

$\xi = (v, \omega) \in \mathbb{R}^6$. $\hat{\xi}\bar{p} = (\omega \times p + v,\ 0)$ = a vector.

Revolute: $\xi = (-\omega \times q,\ \omega) = (q \times \omega,\ \omega)$, $\|\omega\| = 1$, q any point on the axis.

Prismatic: $\xi = (v, 0)$, $\|v\| = 1$; location irrelevant. **Screw:** $\xi = (-\omega \times q + h\omega,\ \omega)$. Axis through origin $\Rightarrow v = 0$. Unit twist: $\|\omega\| = 1$ (or $\|v\| = 1$), **not $\|\xi\| = 1$**.

exp: $\omega = 0$: $\left[\begin{smallmatrix} I & v\theta \\ 0 & 1 \end{smallmatrix}\right]$. $\omega \neq 0$:

$\left[\begin{smallmatrix} e^{\hat{\omega}\theta} & (I - e^{\hat{\omega}\theta})(\omega \times v) + \omega\omega^T v\theta \\ 0 & 1 \end{smallmatrix}\right]$ (**fails for $\omega = 0$**).

Hand shortcut: rotation about axis through q: $e^{\hat{\xi}\theta} = \left[\begin{smallmatrix} R & (I - R)q \\ 0 & 1 \end{smallmatrix}\right]$, i.e. $p \mapsto R(p - q) + q$. Screw: add $h\theta\omega$ to top right. **Check:** a point on the axis maps to itself.

Proof (L7): shift to the closest axis point $g = [I \,|\, \omega \times v]$; $g^{-1}\hat{\xi}g = \left[\begin{smallmatrix} \hat{\omega} & h\omega \\ 0 & 0 \end{smallmatrix}\right]$; $e^{g\hat{\xi}g^{-1}\theta} = ge^{\hat{\xi}\theta}g^{-1}$.

Surjective (P2.9): $R = I$: $v = p/\|p\|, \theta = \|p\|$. Else solve R (P2.5), then $Av = p$, $A = (I - e^{\hat{\omega}\theta})\hat{\omega} + \omega\omega^T\theta$ (invertible for $\theta \neq 0$). Not injective.

se(2): $\hat{\xi} \in \mathbb{R}^{3\times3}$, $\hat{\omega} = \left[\begin{smallmatrix} 0 & -\omega \\ \omega & 0 \end{smallmatrix}\right]$, $v \in \mathbb{R}^2$, $\xi = (v, \omega) \in \mathbb{R}^3$; rotation about q: $v = (\omega q_y, -\omega q_x)$. Every planar motion is a rotation (possibly about a point at ∞).

Screws (Chasles: every motion is a scre

Pitch $h = \frac{\omega^T v}{\|\omega\|^2}$ (∞ if ω=0). Axis $\frac{\omega \times v}{\|\omega\|^2} + \lambda\omega$ (or λv). Magnitude $M = \|\omega\|$ (else $\|v\|$); for $\xi\theta$ with unit ξ, $M = |\theta|$. h=0 rotation, h=∞ translation. Any V^s, V^b has a screw. Thread: n thr/in $\Rightarrow h = \frac{1}{2\pi n}$ in/rad.

Adjoint

$\mathrm{Ad}_g = \left[\begin{smallmatrix} R & \hat{p}R \\ 0 & R \end{smallmatrix}\right]$, $\mathrm{Ad}_g^{-1} = \mathrm{Ad}_{g^{-1}} = \left[\begin{smallmatrix} R^T & -R^T\hat{p} \\ 0 & R^T \end{smallmatrix}\right]$,

$(\mathrm{Ad}_g\xi)^\wedge = g\hat{\xi}g^{-1}$.

Shortcut: revolute $\mathrm{Ad}_g\xi = (-\omega' \times q', \omega')$, $\omega' = R\omega$, $q' = Rq + p$. Prismatic: $(Rv, 0)$. Move the axis, rebuild the twist (trig for planar arms).

Current spatial twist of joint i: $\xi_i' = \mathrm{Ad}_{e^{\hat{\xi}_1\theta_1}\cdots e^{\hat{\xi}_{i-1}\theta_{i-1}}}\xi_i$.

Twist in frame B: $\xi^b = \mathrm{Ad}_{g_{sb}}^{-1}\xi^s$. **Joint i's twist in any link frame $j \geq i$ (incl. tool) is constant**: last joint in tool coords $= \mathrm{Ad}_{g_{st}(0)}^{-1}\xi_n$ for every θ. Read it off the picture.

Rodrigues written out ($v_\theta = 1 - c_\theta$)

$$R = \begin{bmatrix} \omega_1^2 v_\theta + c_\theta & \omega_1\omega_2 v_\theta - \omega_3 s_\theta & \omega_1\omega_3 v_\theta + \omega_2 s_\theta \\ \omega_1\omega_2 v_\theta + \omega_3 s_\theta & \omega_2^2 v_\theta + c_\theta & \omega_2\omega_3 v_\theta - \omega_1 s_\theta \\ \omega_1\omega_3 v_\theta - \omega_2 s_\theta & \omega_2\omega_3 v_\theta + \omega_1 s_\theta & \omega_3^2 v_\theta + c_\theta \end{bmatrix}$$

Equivalently $R = c_\theta I + s_\theta\hat{\omega} + (1 - c_\theta)\omega\omega^T$. $\mathrm{tr}R = 1 + 2c_\theta$.

Handy: $R_z(90°) = \left[\begin{smallmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{smallmatrix}\right]$; 180° about unit ω: $R = 2\omega\omega^T - I$.

$e^{g\hat{\xi}g^{-1}\theta} = ge^{\hat{\xi}\theta}g^{-1}$; $\frac{d}{dt}e^{\hat{\omega}\theta} = \hat{\omega}\dot{\theta}e^{\hat{\omega}\theta}$;

$e^{-\hat{\xi}\theta} = (e^{\hat{\xi}\theta})^{-1}$; $\mathrm{Ad}_{e^{\hat{\xi}\theta}}\xi = \xi$.

Worked: exp of an offset revolute

SCARA joint 2: $\omega = z$, $q = (0, l_1, 0)$:

$e^{\hat{\xi}_2\theta_2} = \left[\begin{smallmatrix} c_2 & -s_2 & 0 & l_1 s_2 \\ s_2 & c_2 & 0 & l_1(1 - c_2) \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{smallmatrix}\right]$. Check: $(0, l_1, 0) \mapsto (0, l_1, 0)$.

Revolute about $+x$ through $(0, a, b)$: $\xi = (0, b, -a, 1, 0, 0)$. About $+y$ through $(a, 0, b)$: $q \times y = (-b, 0, a)$. About $+z$ through $(a, b, 0)$: $q \times z = (b, -a, 0)$.

T/F traps: rotations & twists

- $R_{CA} = R_{BC}^T R_{BA}$ (A wrt B is R_{BA}).
- $\dot{R} \notin so(3)$; $\det\hat{\omega} = 0$; Rodrigues needs $\|\omega\| = 1$.
- exp: so(3)→SO(3), se(3)→SE(3): surjective, not injective. $R = I$ has ∞ exp coords.
- $\theta = \pi$ is *not* a singularity of axis-angle; $\theta = 0$ is.
- Orthogonal with eigs $1, e^{\pm i\phi} \Rightarrow$ SO(3) (det $= \prod\lambda = 1$).
- Euler maps not injective, not global. so(3): abelian under $+$.
- $\hat{\xi}$ bottom-right $= 0$; $\det g = 1$ (translation too).
- $v = -\omega \times q$ only if pitch 0. Unit twist $\neq$ unit norm.
- General exp formula fails for $\omega = 0$.
- Axis through origin: revolute $v = 0$.

Forward kinematics: POE

$\boxed{g_{st}(\theta) = e^{\hat{\xi}_1\theta_1}\cdots e^{\hat{\xi}_n\theta_n} g_{st}(0)}$, twists in S at $\theta = 0$.

1. $g_{st}(0)$: tool origin from dimension lines; R columns = tool axes (parallel gives I). **2.** Each joint: R/P; ω from arrow + **right-hand rule** (drawn arrow wins); simplest q; $\xi = (q \times \omega, \omega)$ or $(v, 0)$. Check $v \perp \omega$. **3.** exps (shortcut). **4.** Multiply.

Figures: z up, y right, **x out of page**; diagonal into the page $= -x$.

Link frame $g_{sl_i}(\theta) = e^{\hat{\xi}_1\theta_1}\cdots e^{\hat{\xi}_i\theta_i} g_{sl_i}(0)$; point on link i: same with $\bar{p}(0)$.

Math: last joint moves first. **Physical order doesn't matter.** $e^A e^B \neq e^{A+B}$ (twists don't commute). $e^{\hat{\xi}\theta} g \neq g$ in general. $\|g\bar{x}\| = \|x\|$ for vectors. $g_{st}(\theta)$ acts on *tool*-coord points; a spatial ref point moves by $e^1\cdots e^n\bar{p}$.

SCARA: $\omega_{1,2,3} = z$,
$q_1 = 0, q_2 = (0, l_1, 0), q_3 = (0, l_1{+}l_2, 0)$:
$\xi_2 = (l_1, 0, 0, 0, 0, 1)$, $\xi_3 = (l_1{+}l_2, 0, 0, 0, 0, 1)$,
$\xi_4 = (0, 0, 1, 0, 0, 0)$, $g_{st}(0) = [I|(0, l_1{+}l_2, l_0)]$.

Elbow (3.11):
$\omega_1{=}z, \omega_{2,3}{=}-x, \omega_4{=}z, \omega_5{=}-x, \omega_6{=}y$;
$q_{1,2} = (0, 0, l_0)$, $q_3 = (0, l_1, l_0)$,
$q_{4,5,6} = (0, l_1{+}l_2, l_0)$.

D-H (course version = relative twists)

$g_{st} = g_{sl_1}(\theta_1) g_{l_1l_2}(\theta_2)\cdots g_{l_{n-1}l_n}(\theta_n) g_{l_nt}$,

$g_{l_{i-1}l_i}(\theta_i) = e^{\hat{\xi}_{i-1,i}\theta_i} g_{l_{i-1}l_i}(0)$.

$\xi_{i-1,i}$ = joint i twist in L_{i-1} coords (constant at any config). $L_0{=}S \Rightarrow \xi_{0,1}{=}\xi_1$.

$\boxed{\xi_i = \mathrm{Ad}_{g_{sl_{i-1}}(0)}\xi_{i-1,i}}$. Frames parallel to S:

revolute $\xi_{i-1,i} = (-\omega_i \times (q_i - o_{i-1}), \omega_i)$; prismatic unchanged. $g_{l_nt} \neq g_{st}(0)$.

If L_i is on axis i: $g_{l_{i-1}l_i}(\theta_i) = \begin{bmatrix} e^{\hat{\omega}_i\theta_i} & p_{l_{i-1}l_i}(0) \\ 0 & 1 \end{bmatrix}$ (spins in place). Prismatic: $[R(0)\,|\,p(0) + v\theta_i]$.

Inverse kinematics

$e^1\cdots e^n = g_1 := g_d g_{st}(0)^{-1}$. Max #solutions is a property of the robot + g_d, **not the method**.

Tricks: (1) point on a revolute axis: $e^{\hat{\xi}\theta}\bar{p} = \bar{p}$ (kills exps). (2) Subtract a point fixed by the first joints, take the norm (distance preserved).

Planar RR: $\cos\theta_2 = \frac{r^2 - l_1^2 - l_2^2}{2l_1l_2}$ $(\pm)$,
$\theta_1 = \mathrm{atan2}(y, x) - \mathrm{atan2}(l_2s_2, l_1 + l_2c_2)$. 2/1/0 solutions. Redundant (RRRR in SE(2), 7-DOF in SE(3)): ∞.

$u = p - r$, $v = q - r$ (r on axis),
$u' = u - \omega\omega^T u$, $v' = v - \omega\omega^T v$.

SP1 $e^{\hat{\xi}\theta}p = q$: needs $\omega^T u = \omega^T v$ **and** $\|u'\| = \|v'\|$ **(not just $p \neq q$).**
$\theta = \mathrm{atan2}(\omega^T(u' \times v'), u'^T v')$. 1 solution ($\infty$ if $u' = 0$).

SP2 $e^{\hat{\xi}_1\theta_1} e^{\hat{\xi}_2\theta_2} p = q$, axes meet at r:
$z = \alpha\omega_1 + \beta\omega_2 + \gamma(\omega_1 \times \omega_2)$,
$\alpha = \frac{(\omega_1^T\omega_2)\omega_2^T u - \omega_1^T v}{(\omega_1^T\omega_2)^2 - 1}$, $\beta = \frac{(\omega_1^T\omega_2)\omega_1^T v - \omega_2^T u}{(\omega_1^T\omega_2)^2 - 1}$,
$\gamma^2 = \frac{\|u\|^2 - \alpha^2 - \beta^2 - 2\alpha\beta\omega_1^T\omega_2}{\|\omega_1 \times \omega_2\|^2}$.
$\perp$ axes: $\alpha = \omega_1^T v$, $\beta = \omega_2^T u$, $\gamma^2 = \|u\|^2 - \alpha^2 - \beta^2$.
$c = z + r$. SP1: $p \to c$ about ω_2 gives θ_2; $c \to q$ about ω_1 gives θ_1. 2/1/0 solutions ($\gamma^2 >, =, < 0$).

SP3 $\|q - e^{\hat{\xi}\theta}p\| = \delta$: $\delta'^2 = \delta^2 - |\omega^T(p - q)|^2$,
$\theta_0 = \mathrm{atan2}(\omega^T(u' \times v'), u'^T v')$,
$\theta = \theta_0 \pm \cos^{-1}\frac{\|u'\|^2 + \|v'\|^2 - \delta'^2}{2\|u'\|\|v'\|}$. 2/1/0 solutions.

Elbow (8 solutions): θ_3: apply to wrist ctr p_w, subtract p_b (axes 1,2 meet), norm:
$\|e^3p_w - p_b\| = \|g_1p_w - p_b\|$ (SP3, 2). $\theta_{1,2}$:
$e^1e^2(e^3p_w) = g_1p_w$ (SP2, 2). $\theta_{4,5}$:
$g_2 = e^{-3}e^{-2}e^{-1}g_1$ on a pt of axis 6 off axes 4,5 (SP2, 2). θ_6: SP1 on a pt off axis 6 (1).

SCARA: $\theta_4 = z - l_0$; θ_2 SP3; θ_1 SP1; θ_3 SP1; $\theta_1 + \theta_2 + \theta_3 = \phi$.

Facts: R+P joints on the same axis commute (swap gives the same IK). Given the other joints, one revolute $\theta \in [0, 2\pi)$ is unique. Fully stretched: the elbow pair merges, but base/wrist flips remain.

Rigid body velocity

$\dot{R}R^T$, $R^T\dot{R} \in so(3)$ **but $\dot{R} \notin so(3)$.** $\hat{\omega}^s = \dot{R}R^T$, $\hat{\omega}^b = R^T\dot{R}$, $\omega^s = R\omega^b$.
$\hat{V}^s = \dot{g}g^{-1}$: $V^s = (-\dot{R}R^Tp + \dot{p},\ (\dot{R}R^T)^\vee)$.
$\hat{V}^b = g^{-1}\dot{g}$: $V^b = (R^T\dot{p},\ (R^T\dot{R})^\vee)$.

$\boxed{V^s = \mathrm{Ad}_g V^b}$, $V^b_{ab} = -V^s_{ba} = -\mathrm{Ad}_{g_{ba}} V^b_{ba}$; V^b independent of the inertial frame.

v^s = velocity of the body point at A's origin (not B's origin). v^b = vel. of B's origin in B coords (**can be $\neq 0$**).
Point: $v_{q_a} = \omega^s \times q_a + v^s$; $v_{q_b} = \omega^b \times q_b + v^b$ (bottom 0).

Rotation about a fixed axis (ω, q) at rate $\dot{\theta}$: $V^s = \xi\dot{\theta} = (-\omega \times q, \omega)\dot{\theta}$; add v_{trans} to the top for a screw. Constant V^s: $g(t) = e^{\hat{V}^s t}g(0)$; constant V^b: $g(t) = g(0)e^{\hat{V}^b t}$.

Wrenches

$F = (f, \tau)$ at a frame's origin, in its coords. Power $V^b \cdot F_b = v \cdot f + \omega \cdot \tau$. **Frame change via Ad^T, not g**: $F_c = \mathrm{Ad}^T_{g_{bc}} F_b$, $\mathrm{Ad}^T_g = \begin{bmatrix} R^T & 0 \\ -R^T\hat{p}\,R^T & R^T \end{bmatrix}$:
$f_c = R^T f_b$, $\tau_c = R^T(\tau_b - p \times f_b)$. **Net wrench depends on the reference point.**

Manipulator Jacobian

$\boxed{J^s = [\xi_1\ \xi_2'\cdots\xi_n']}$, $\xi_i' = \mathrm{Ad}_{e^1\cdots e^{i-1}}\xi_i$, $V^s = J^s\dot{\theta}$.
Col i depends on $\theta_{1..i-1}$; **no θ_n**.
$J^s(0) = [\xi_1\cdots\xi_n]$.
$J^b = [\xi_1^\dagger\cdots\xi_n^\dagger]$, $\xi_i^\dagger = \mathrm{Ad}^{-1}_{e^i\cdots e^n g_{st}(0)}\xi_i$ = twist in tool coords. Col i depends on $\theta_{i+1..n}$; **no θ_1**; **moving joint i doesn't change col i.** Last col constant. $J^s = \mathrm{Ad}_{g_{st}}J^b$. $J^b(0)$: axes re-measured from the tool origin.

Column by hand: $\omega_i' = R\omega_i$, q_i' = where joint i's axis point moved to (apply earlier joints), $\xi_i' = (q_i' \times \omega_i', \omega_i')$.

Statics: $\boxed{\tau = J^T F}$: $(J^b_{st})^T F_{tool}$ or $(J^s_{st})^T F_s$.
Transpose! Wrench on link k: use J_{sl_k} (cols $> k$ zero). Resisting an external F: $-J^T F$.

Singular (rank drop): 2 collinear revolute axes (wrist $\theta_5 = 0$ lines up 4 and 6); 3 parallel coplanar R; 4 R axes through one point; 4 coplanar R; 6 R meeting one line; P $\perp$ plane of 2 parallel coplanar R.

Sanity checks

- $v \perp \omega$ for revolute; prismatic $\omega = 0$; (v, ω) order: v on top.
- Axis point fixed by its own exp; $\det R = 1$, $R^TR = I$; $\mathrm{tr}R = 1 + 2\cos\theta$.
- Twist $\times$ point gives bottom 0; $g\times$ point gives bottom 1.
- Planar check: speed $= |\omega| \cdot$(distance to axis), direction tangent.

Exam recipes

Q6 (FK): mark S, T; $g_{st}(0)$; per joint: R/P, ω by RHR on the drawn arrow, simplest q, ξ; check $v \perp \omega$; D-H: re-measure q from L_{i-1}; $g_{l_{i-1}l_i}$ multiplied out 4×4; tool-frame twist of the last joint = read it from T (constant). One reason per line.

Q7 (vel/J/F): twists $\to J^s$ columns (move q, ω by earlier joints) $\to$ at $\theta = 0$, $J = [\xi_i] \to$ $v_q = \omega \times q + v \to$ single joint moving: $V^s = \xi_k'\dot{\theta}_k$ $\to \tau = J^T F$ (body wrench gives J^b).

Worked: J^s column 3

J1 z@0, J2 x@0, J3 x@$(0, 0, l_1)$.
$\omega_3' = R_z(\theta_1)R_x(\theta_2)x = (c_1, s_1, 0)$; $q_3' = R_z(\theta_1)(0, -l_1s_2, l_1c_2) = (l_1s_1s_2, -l_1c_1s_2, l_1c_2)$;
$v_3' = q_3' \times \omega_3' = (-l_1s_1c_2,\ l_1c_1c_2,\ l_1s_2)$. Col 2: $(0, 0, 0, c_1, s_1, 0)$. Col 1: $(0, 0, 0, 0, 0, 1)$.
Tool at $(0, 0, L)$, $L = l_1 + l_2$: $J^b(0)$ cols: $(0, 0, 0, 0, 0, 1)$, $(0, -L, 0, 1, 0, 0)$, $(0, -l_2, 0, 1, 0, 0)$.
$\tau = J^{bT}F_b = (\tau_z,\ \tau_x - Lf_y,\ \tau_x - l_2f_y)$.

Worked: SP2 ($\omega_1 = z, \omega_2 = x$, $r = 0$)

$p = (0, 1, 0)$, $q = (-\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}})$: $\alpha = q_z$,
$\beta = p_x = 0$, $\gamma^2 = 1 - \frac{1}{2}$; $c = (0, \pm\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})$
$(\omega_1 \times \omega_2 = y)$. $\theta_2 = 45°/135°$, $\theta_1 = \pm 90°$.

Worked: velocity

Body at $(2, 0, 0)$, spinning about its z at π:
$V^s = (-\omega \times q, \omega) = (0, -2\pi, 0, 0, 0, \pi)$. Point $q_a = (3, 0, 0)$: $v = \omega \times q_a + v^s = (0, \pi, 0)$. Plus $3\,\hat{z}$ translation: $v_z^s = 3$, $h = 3/\pi$.

T/F traps: FK, IK, velocity, J

- $\xi_3' = \mathrm{Ad}_{e^1e^2}\xi_3$ = current spatial twist. $e^{\hat{\xi}\theta}g \neq g$.
- $\|g(\bar{p} - \bar{q})\| = \|\bar{p} - \bar{q}\|$. $g_{l_0l_1} = e^{\hat{\xi}_1\theta_1} g_{l_0l_1}(0)$ (since $L_0 = S$).
- D-H twist definable at any config (constant in L_{i-1}).
- Max #IK solutions is independent of the method. RRRR planar: ∞.
- SP1 not solvable for arbitrary $p \neq q$. Point on axis is fixed.
- Stretched arm: not necessarily 1 solution. R/P on one axis commute.
- Every V^s, V^b has a screw. $v^s \neq$ velocity of B's origin; v^b can be $\neq 0$.
- Wrench: Ad^T, not g; depends on the reference point.
- J^s indep. of θ_n; J^b indep. of θ_1 (not θ_n); col i of J^b indep. of θ_i.
- $\tau = J^T F$ (transpose; the right link's Jacobian).